

PACT: Compliance Reporting in PPE Detection based on Pose-Anchored Body Region Assignment

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Abstract—Automated systems that detect personal protective equipment (PPE) in images typically report only scene-level presence, failing to assign each item to a specific worker and preventing per-worker compliance checks required by safety regulations. This paper addresses this gap with Pose-Anchored Compliance Tracker (PACT), a three-stage pipeline. First, a YOLO26L detector locates PPE items and a YOLOv8x-pose model identifies workers and body keypoints. Second, a pose-anchored matching module attributes each detected PPE instance to the corresponding person. Third, a rule-based reporter computes fractional compliance scores and severity-tiered violation notices for each worker. We evaluate on three public datasets: construction-site CHV, medical PPE CPPE-5, and industrial SH17. PACT achieves mean average precision at IoU 0.50 (mAP₅₀) of 0.923 on CHV and 0.652 on SH17, while per-person assignment accuracy reaches 0.938 on CHV and 0.892 on SH17. These results demonstrate that pose estimation enables reliable person-level PPE attribution and that assignment accuracy is a measurable target for compliance reporting.

Index Terms—PPE, Compliance reporting, Pose Estimation, YOLOv8-pose, YOLO26

person scenes. Some works incorporate pose estimation to filter false positives [8], [9], [10] but pose is used strictly as a preprocessing step. None of these works define PPE-to-person assignment as an explicit task or produce per-person compliance scores. Vukicevic et al. [4] identify this as an open problem, noting that practical compliance systems require joint handling of human identification, pose estimation, and object recognition.

This work presents Pose-Anchored Compliance Tracker (PACT), a three-stage pipeline consisting of parallel YOLO26L [11] detection and YOLOv8x-pose estimation [12], followed by a pose-anchored PPE-to-person assignment module and a rule-based per-person compliance reporting mechanism. PACT formalizes PPE-to-person assignment as an explicit evaluation task by introducing a dedicated assignment accuracy metric for multi-person scenarios. A multi-stage evaluation protocol independently quantifies person detection, PPE detection, and assignment accuracy across 10,458 images.

I. INTRODUCTION

Occupational safety regulations including Occupational Safety and Health Administration (OSHA) 29 CFR 1926 [1], ANSI/ISEA 107-2020 [2], and ISO 45001 [3] mandate the use of personal protective equipment (PPE) across construction, manufacturing, and healthcare environments. Non-compliance contributes to preventable workplace injuries, while manual inspection is neither scalable nor consistently reliable in large and dynamic work sites [4]. Automated reporting through computer vision has therefore emerged as a practical alternative for continuous PPE verification.

Current approaches treat PPE compliance as an object detection problem, applying YOLO-based detectors to construction CHV [5], manufacturing SH17 [6], and medical CPPE-5 [7] datasets. Detection-only systems produce PPE bounding boxes without associating each item to a specific worker, making per-person compliance evaluation impossible in multi-

II. RELATED WORK

A. PPE Detection and Pose-Assisted Compliance

Single-stage YOLO detectors have become the dominant approach for PPE compliance due to their accuracy and low latency [4]. Wang et al. introduced the CHV dataset and benchmarked eight YOLO variants, with YOLOv5x achieving 0.866 mAP [5]; Kisaeezra et al. reported 0.958 mAP and 70.4 FPS on helmet detection using YOLOv5 variants [16]. Gallo et al. demonstrated real-time edge deployment using lightweight CNNs [13]. Two-stage detectors have also been explored: Ahmed et al. achieved 0.96 mAP with Faster R-CNN on eight PPE classes but were limited to single-person evaluation [15]. Across these works, detection is treated as a class-presence problem: the system confirms whether a PPE category exists in the scene without attributing items to specific workers.

TABLE I
COMPARISON OF REPRESENTATIVE PPE COMPLIANCE METHODS (WITH DATASETS)

Method	Publication Year	Detector	Domain	Dataset	Pose Usage	Per-Person Compliance	Severity Report	Performance
Wang et al. [5]	2021	YOLOv5	Construction	GDUTâ€™Hardhat (custom)	None	No	–	No mAP 0.866 (YOLOv5x)
Gong et al. [10]	2021	YOLOv3 + ResNet50	Offshore	Custom offshore platform	ROI	Yes	Binary	No Helmet 94.8% / Workwear 95.4%
Gallo et al. [13]	2022	YOLOv4	Industrial	PICTOR	None	No	–	No mAP 0.863 / 6.8 FPS
Vukicevic et al. [14]	2022	YOLOv5	Industrial	Custom headâ€™mounted images	ROI	Yes	Binary	No Precision 0.920 / Recall 0.611
Ahmed et al. [15]	2023	Faster Râ€™CNN	General	Mixed public + webâ€™scraped	None	Single	–	No mAP 0.960
Kisaezhe et al. [16]	2023	YOLOv5	Construction	Construction images (Roboflow)	None	No	–	No mAP 0.958 / 70.4 FPS
Monnikhof et al. [8]	2023	YOLOv7 + pose model	Custom	Custom pipeline recordings	Filter	No	–	No FP reduction 0.22
Imam et al. [9]	2024	YOLOv8-pose + YOLOv8l	Mining	Selfâ€™collected underground mine	Filter	Yes	Binary	No Pose mAP 0.90 / PPE mAP 0.97
Wen et al. [17]	2024	YOLO + VQA	Construction	Custom construction VQA	None	No	Scene	No Not reported

Pose Usage: None = not used; ROI = keypoint-defined detection regions; Filter = false-positive suppression. **Per-Person:** Single = evaluated on single-person images only.
Compliance Type: Binary = pass/fail per person, Scene = scene-level label, Fractional = continuous per-person score, – = not reported. **Severity Report:** Yes = severity-tiered corrective action report provided. All mAP values are at IoU 0.50 unless stated otherwise.

Recent works incorporate pose estimation to reduce false positives. Monnikhof et al. used YOLO-pose keypoints as anatomical constraints to filter implausible detections, reducing false positives by 0.22 [8]. Imam et al. applied YOLOv8-pose to discard unlinked PPE detections in mining environments [9], and Gong et al. defined keypoint-based ROIs before PPE classification [10]. Despite improved precision, pose is used solely as a post-detection filter in all cases; none formalize PPE-to-person assignment as an explicit task or evaluate per-person attribution in multi-worker scenes. Vukicevic et al. identify this as an open problem, noting that practical systems must jointly address detection, pose estimation, and assignment [4]. This gap is the primary motivation for the proposed framework.

B. Compliance Reporting

Earlier PPE compliance systems produce a single binary safety label for the entire scene [17], which is insufficient for multi-person environments where individual accountability is required. Systems that check compliance separately for each

detected person represent a step toward per-worker granularity [18], and pipelines that identify missing PPE at the individual level provide the specific information needed for worker records [14]. However, none of these works formalize a per-person compliance score or produce severity-tiered corrective action items linked to worker identifiers, leaving targeted safety audits and longitudinal compliance tracking unsupported.

Table I summarizes the key characteristics of representative prior systems. The comparison covers pose integration strategy, per-person PPE attribution, compliance score granularity, severity-tiered reporting, and dataset domain. Two dimensions distinguish PACT from these prior systems: pose is used for assignment rather than filtering, and the system produces fractional per-person compliance scores with severity-tiered corrective action items rather than binary pass/fail outputs or scene-level labels.

PACT: Pose-Anchored Compliance Tracker System Pipeline

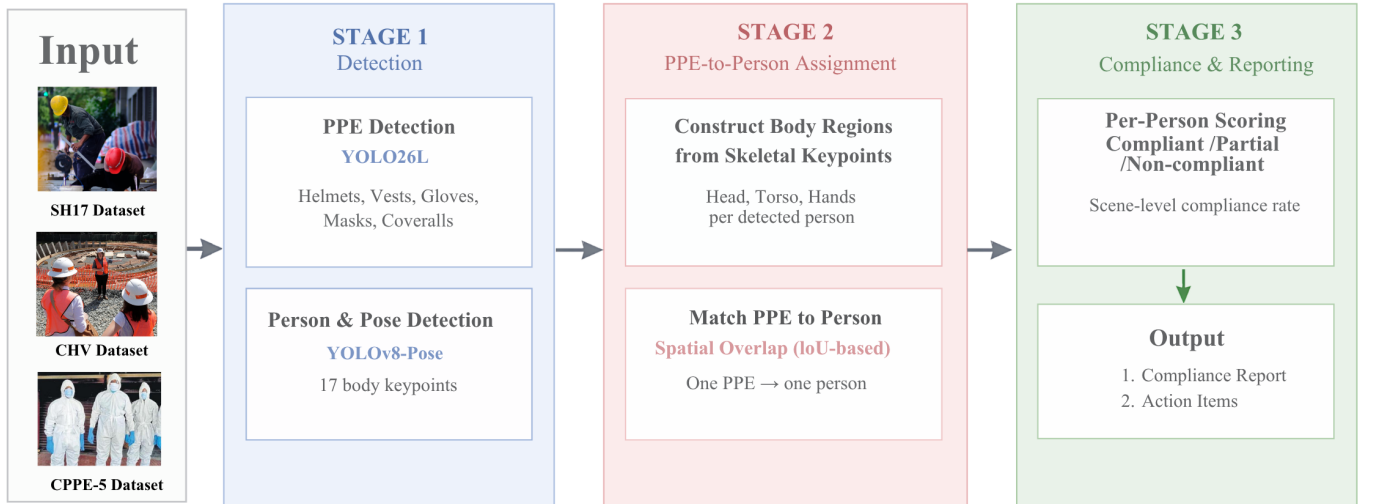


Fig. 1. Overview of the PACT pipeline showing parallel PPE detection and pose estimation, followed by pose-anchored assignment and per-person compliance reporting.

III. METHOD

A. Datasets

Three publicly available datasets covering distinct deployment domains are used for evaluation. CHV [5] contains 1,330 construction site images annotated across six classes (person, safety vest, and four helmet colors), evaluated under the original split (1,064/133/133 train/val/test). CPPE-5 [7] contains 1,029 images across five medical PPE classes (coverall, face shield, gloves, goggles, mask), evaluated under the original split (1,000 train / 29 test); the dataset contains no person-class annotations. SH17 [6] contains 8,099 industrial images annotated across 17 classes covering both body parts and PPE items, it is evaluated under the original split (6,479 train / 1,620 validation, used as test). All images are resized to 640×640 pixels prior to inference.

B. System Overview

PACT processes a single input image through three sequential stages, as illustrated in Fig. 1 and exemplified by Table II. In Stage 1 (Parallel Inference), a YOLO26L object detector and a YOLOv8x-pose model run simultaneously on the same input. YOLO26L produces a set of PPE bounding boxes, each labeled with a class and confidence score. YOLOv8x-pose independently localizes each person in the scene and extracts a skeleton of 17 body keypoints per person in COCO format.

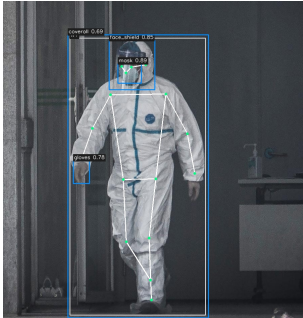
In Stage 2 (Pose-Anchored Assignment), the PACT assignment module takes the PPE detections from YOLO26L and the skeletal keypoints from YOLOv8x-pose as input. For each detected PPE item, the module constructs anatomical body regions from the keypoints of each person and determines which person the item most plausibly belongs to using spatial overlap reasoning. This produces a set of per-person PPE attributions rather than a scene-level list of detected items.

In Stage 3 (Compliance Evaluation and Reporting), a rule-based module evaluates each worker's attributed PPE against a dataset-specific required set, computes a fractional compliance score per person, and generates a structured violation report with severity-tiered corrective action items.

C. PPE Detection Module

PPE detection uses YOLO26L [11], a large-scale YOLO26 variant pretrained on COCO and fine-tuned per dataset. The model contains 24.75M parameters at 86.1 GFLOPs. Training uses AdamW ($\eta_0 = 1 \times 10^{-3}$, weight decay 5×10^{-4}), batch size 8, up to 150 epochs with early stopping at patience 20, and Automatic Mixed Precision (AMP). Augmentations include mosaic, horizontal flip, HSV jitter, random scale and translation, RandAugment, and random erasing. Each detection carries a bounding box, class label, and confidence score. Sub-

TABLE II
QUALITATIVE RESULTS OF THE PROPOSED PACT FRAMEWORK

Dataset	Raw	Pose	Pose + PPE Detection	Report
CHV				Context: Industrial Workers: 2 Compliance: - Worker 1 (66% compliant) - Worker 2 (33% compliant) Worker Status: - W1: Helmet [OK] Vest [OK] Gloves [X] - W2: Helmet [X] Vest [OK] Gloves [X] Action items: [URGENT] Provide Helmet: Worker 2 [ADVISORY] Provide Gloves: Worker 1, Worker 2
CPPE-5				Context: Healthcare Workers: 1 Compliance: - Worker 1 (100% compliant) Worker Status: - W1: Coverall [OK] Mask [OK] Face Shield [OK] Gloves [OK] Action items:
SH17				Context: Industrial Workers: 2 Compliance: - Worker 1 (100% compliant) - Worker 2 (66% compliant) Worker Status: - W1: Vest [OK] Helmet [OK] Gloves [OK] - W2: Vest [X] Helmet [OK] Gloves [OK] Action items: [URGENT] Provide Vest: Worker 2

threshold and person-class detections are discarded, as person localization is delegated to the pose module.

D. Person and Pose Detection Module

Person localization and keypoint extraction use YOLOv8x-pose [12], applied in inference mode without fine-tuning. It jointly detects persons and regresses 17 COCO-format skeletal keypoints per person, each with normalized coordinates and a visibility confidence score. Since undetected persons cannot receive PPE attribution, detection rate directly bounds assignment coverage. The extracted keypoints serve as geometric anchors for the assignment stage.

E. PPE to Person Assignment

The assignment module is the core contribution of PACT. It attributes each PPE detection to a specific worker using pose-anchored spatial reasoning, operating in two steps.

1) *Body Region Construction*: For each detected person, a set of anatomical body regions is constructed from their visible keypoints. Each PPE class maps to one or more regions based on its expected wearing location: helmets map to the head, vests to the torso, and gloves to the hands. Each region is defined as the bounding box enclosing its associated keypoints, expanded by a margin of δ times the region bounding box dimensions. A region is only constructed if at least two of its keypoints exceed a visibility threshold of 0.3, suppressing regions based on poorly localized estimates.

2) *IoU-Based Matching*: Each PPE detection is assigned to the person whose anatomical body region yields the highest IoU. An assignment is accepted if $IoU > \tau_a = 0.10$. This permissive threshold is intentionally lower than the standard 0.50 because overlap is computed against compact anatomical anchors rather than identical object boxes. Since wearable items (e.g., helmets, gloves) occupy only a fraction of their corresponding body regions, a higher threshold would reject valid assignments. In multi-person scenes, each item is attributed to at most one worker; if multiple detections of the same class match one person, only the highest-confidence instance is retained.

F. Compliance Scoring and Reporting

After assigning detected PPE items to individual workers, the compliance evaluation module computes a per-person compliance score defined as the ratio of confirmed worn items to the total required equipment for that deployment context. Required PPE sets are defined per dataset based on applicable regulations: helmet and vest for CHV (OSHA 29 CFR 1926.100, ISO 20471); coverall, mask, and gloves for CPPE-5 (WHO/CDC infection control guidelines); and helmet, safety-vest, and gloves for SH17 (OSHA 29 CFR 1926, ISO 45001). A score of 1.0 indicates full compliance, while partial compliance is tracked separately from a binary pass/fail classification. The scene-level compliance rate is computed as the mean of all per-person scores.

For each worker, every missing required item is converted into a corrective action item. Each action item is prefixed with

[URGENT] or [ADVISORY] based on a static severity tier assigned to the PPE class. Four tiers are defined: *Critical* (helmet, hard hat), *High* (vest, coverall, mask), *Medium* (gloves, face shield, goggles), and *Low* (shoes, protective suits). A missing item at the Critical or High tier receives [URGENT]; Medium or Low receives [ADVISORY]. Formally, for a missing class c ,

$$u(c) = \begin{cases} \text{URGENT} & \tau(c) \in \{\text{CRITICAL, HIGH}\} \\ \text{ADVISORY} & \text{otherwise} \end{cases} \quad (1)$$

where $\tau: \mathcal{C} \rightarrow \{\text{CRITICAL, HIGH, MEDIUM, LOW}\}$ is the severity mapping. Action items are sorted by descending severity before output, so that the highest-risk violations appear first.

The reporting module merges these results into a structured output containing per-person compliance scores, the list of identified missing items, a scene-level summary, and the severity-sorted corrective action items grouped by worker identifier.

G. Implementation Details

YOLO26L is fine-tuned independently on each dataset split from COCO-pretrained weights. All experiments are conducted on an NVIDIA GeForce RTX 4050 Laptop GPU (6 GB VRAM) under CUDA 11.8, PyTorch 2.7, and Ultralytics 8.4.31.

For person localization and keypoint extraction, YOLOv8x-pose is applied in inference mode without fine-tuning on any target dataset. The model produces 17 skeletal keypoints per detected person in COCO body format, represented as (x, y, v) where v denotes visibility confidence. PACT assignment uses an anchor-overlap threshold $\tau_a = 0.10$, keypoint visibility threshold $\tau_v = 0.3$, and body region expansion margin $\delta = 0.10$ (10% of the enclosing keypoint bounding box dimensions) across all datasets.

H. Evaluation Metrics

The evaluation framework is structured across three stages corresponding to the PACT pipeline. Stage 1 covers parallel inference, with person detection and PPE detection evaluated separately. For Stage 1a, YOLOv8x-pose is evaluated against ground-truth person bounding boxes at IoU threshold 0.50, reporting precision, recall, F1-score, and TP/FP/FN counts. For Stage 1b, YOLO26L is evaluated against ground-truth PPE bounding boxes at IoU threshold 0.50, reporting mAP50 and mAP50-95 for comparison with prior benchmarks [5]–[7].

Stage 2 evaluates PPE-to-person assignment accuracy on multi-person frames containing at least two ground-truth persons. CHV and SH17 do not provide explicit person-PPE relationship labels; therefore, ground-truth pairs are derived geometrically. Each ground-truth PPE box is paired with the ground-truth person box that covers the largest fraction of the PPE box, and PPE boxes with no meaningful person overlap are excluded. A matched detection ($IoU \geq 0.50$ with its ground-truth PPE item) is counted as correctly assigned if

PACT attributed it to the same derived ground-truth person. CPPE-5 is excluded from this Stage 2 metric because it has no person-class ground-truth boxes. Let A denote the number of correctly assigned detections and T the total number of matched detections across all qualifying frames. Assignment accuracy is defined as:

$$\text{Acc}_{\text{assign}} = \frac{A}{T} \quad (2)$$

End-to-end latency is measured on the CPPE-5 test set by timing the PPE detection stage, the pose detection stage, and the anchor construction and compliance stage independently. Three warmup images are discarded before measurement. Latency is reported as the mean over all 29 test images.

Stage 3 does not introduce a dedicated metric; compliance scores and violation reports are evaluated qualitatively through the structured outputs.

IV. RESULTS AND DISCUSSION

A. Performance on Stage 1

Table III shows person detection results for YOLOv8x-pose. CHV achieves F1 of 0.850 with precision 0.907 and recall 0.800. SH17 achieves the highest F1 of 0.887 with precision 0.944, though 448 persons remain undetected across 1,620 images, indicating that worker density increases missed detections despite high per-detection accuracy. CPPE-5 reaches F1 of only 0.483, with low precision (0.396). Displayed in Table IV, CHV mAP50 achieves 0.923 and mAP50-95 of 0.548, while CPPE-5 and SH17 achieve 0.765 and 0.652 respectively for mAP50.

TABLE III
PERSON DETECTION PERFORMANCE OF YOLOv8x-POSE

Dataset	Precision	Recall	F1	TP	FP	FN
CHV	0.907	0.800	0.850	360	37	90
SH17	0.944	0.836	0.887	2286	136	448
CPPE-5	0.396	0.618	0.483	21	32	13

TP: True Positive, FP: False Positive, FN: False Negative

TABLE IV
OVERALL PPE DETECTION PERFORMANCE OF YOLO26L

Dataset	Test Images	mAP50	mAP50-95	Precision	Recall
CHV	133	0.923	0.548	0.926	0.859
CPPE-5	29	0.765	0.527	0.812	0.698
SH17	1,620	0.652	0.432	0.746	0.590

Fig. 2 illustrates representative detection failures across datasets. On CHV and SH17, safety vests are missed because side-view perspectives reduce the visible area of the vest below the detector’s confidence threshold. On CPPE-5, false detections occur in background regions for mask and coverall classes, indicating that the model activates on background patches with color similar to the typical mask and coverall in the dataset.

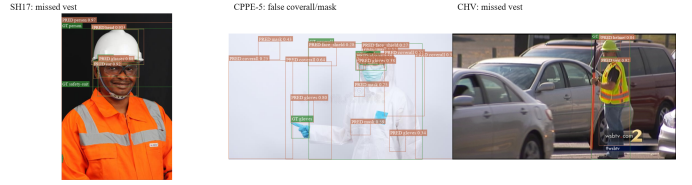


Fig. 2. Representative failure cases across datasets with explicit bounding-box callouts. Green bounding boxes are the ground truth, while the red bounding boxes are the predictions. CHV and SH17 show missed vest detections due to occlusion and side-view perspectives, while CPPE-5 exhibits false positives caused by background similarity.

B. Performance on Stage 2

The accuracy of PPE-to-person assignment, shown in Table V, correlates with scene complexity. CHV achieves the highest accuracy (0.938) because construction scenes typically contain fewer overlapping workers and larger inter-person distances, which reduces ambiguity in body region IoU matching. SH17 yields a lower accuracy of 0.892, reflecting denser worker configurations where bounding boxes from adjacent persons often overlap and increase the probability of misassignment. CPPE-5 is not included in Table V because its annotations do not contain person boxes, making person-PPE assignment accuracy non-comparable without a derived proxy protocol.

TABLE V
PPE-TO-PERSON ASSIGNMENT ACCURACY

Dataset	Correct	Total	Accuracy
CHV	408	435	0.938
SH17	521	584	0.892

CPPE-5 excluded because person ground-truth boxes are unavailable.

C. Runtime Analysis

TABLE VI
MEAN PIPELINE LATENCY ACROSS ALL TEST SETS

Dataset	PPE det. (ms)	Pose det. (ms)	Anchor+C (ms)	Total (ms) / FPS
CPPE-5 ($n = 29$)	29.80	43.59	0.41	73.93 / 13.53
CHV ($n = 133$)	33.28	46.64	0.79	80.86 / 12.37
SH17 ($n = 1,620$)	29.12	62.19	0.39	91.83 / 10.89
Overall ($n = 1,782$)	29.44	60.73	0.42	90.72 / 11.02

Report gen.: 0.13 ms (<1%). GPU: NVIDIA RTX 4050 (6 GB).

Anchor+C = anchor construction + compliance.

Table VI reports mean latency per pipeline stage measured across all three test sets ($n = 1,782$ images), with each stage timed independently. The dominant bottleneck is pose detection (YOLOv8x-pose) at 60.73 ms (67% of total), followed by PPE detection (YOLO26L) at 29.44 ms (32%). Anchor construction and compliance evaluation account for only 0.42 ms (<1%), confirming that the assignment logic imposes negligible overhead relative to neural network inference. The mean end-to-end latency of 90.72 ms corresponds to 11.02 FPS, below the 25 FPS threshold commonly required for real-time video processing. The system is therefore suited

for batch or near-real-time inspection scenarios rather than continuous video monitoring.

D. Ablation Study

We perform a sensitivity analysis on the PPE-to-person assignment module by sweeping τ_a and δ across 20 configurations. Table VII presents the assignment accuracy for both CHV and SH17 datasets.

TABLE VII
ASSIGNMENT ACCURACY ABLATION: $\tau_a \times \delta$ GRID ($\tau_v = 0.30$ FIXED)

CHV (94 multi-person images)				
$\tau_a \backslash \delta$	0.10	0.15	0.20	0.25
0.10	0.9844	0.9820	0.9821	0.9795
0.15	0.9842	0.9844	0.9819	0.9793
0.20	0.9839	0.9840	0.9843	0.9791

SH17 (582 multi-person images)				
$\tau_a \backslash \delta$	0.10	0.15	0.20	0.25
0.10	0.9723	0.9734	0.9632	0.9614
0.15	0.9598	0.9668	0.9692	0.9617
0.20	0.9557	0.9634	0.9636	0.9659

Default: $\tau_a = 0.10, \delta = 0.10$ (bold). CHV: 0.977–0.987. SH17: 0.946–0.974.

Table VII demonstrates PACT’s robustness to parameter variations. On CHV, an accuracy spread below 0.01 reflects stable assignment under clear geometric separation between workers. On the denser SH17, accuracy remains above 0.946 despite increasing δ , suggesting that the default parameters ($\tau_a = 0.10, \delta = 0.10$) generalize well across domains without intensive tuning.

E. Limitations

Several limitations exist in the proposed PACT pipeline. Detection errors in Stage 1 directly propagate to the report, and the assignment module remains susceptible to misattribution in dense scenes. Furthermore, PACT’s generalizability to medical domains is partially unverified due to the lack of person-class annotations in CPPE-5. Additionally, following standard PPE benchmarks [5]–[7], results are reported as point estimates; however, we recognize that the limited size of the CPPE-5 test set may introduce variance in single-point mAP estimates. Future work will address these gaps by performing manual annotation of a CPPE-5 subset for healthcare validation, incorporating statistical variance reporting, and developing temporal tracking to maintain compliance states.

V. CONCLUSION

Detection-only PPE compliance systems cannot attribute equipment items to specific workers in multi-person scenes, limiting their utility in real deployments. PACT addresses this by combining YOLO26L-based PPE detection, YOLOv8x-pose-based person localization, and pose-anchored body region IoU assignment, with a rule-based module generating per-person compliance scores and severity-tiered reports. Evaluated across construction, medical, and industrial datasets under

a three-stage protocol, PACT achieves mAP50 of 0.923 on CHV and 0.652 on SH17, and assignment accuracy of 0.938 on CHV and 0.892 on SH17 for datasets with person ground truth. End-to-end latency of 140.87 ms (7.1 FPS) positions PACT for batch and near-real-time inspection scenarios. Future work will address person detection coverage in crowded and occluded scenes such as CPPE-5, and extend the framework to video-based temporal tracking.

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